

Initial Project Planning Template

Date:	3 October 2026	Team ID:	unm1611MADHANRAJ
Project Name:	AI FitTrack API	Maximum Marks:	5 Marks

Product Backlog, Sprint Schedule, and Estimation:

The agile backlog below details the sprint allocation, user stories, effort estimation in story points, priorities, and timelines for developing and validating AI FitTrack API.

Sprint	Epic / Module	User Story / Technical Task	Points	Priority	Assignee	Timeline
Sprint 1	Scaffolding & Setup	Project init, Express setup, MongoDB Atlas cluster connection, Gemini API key.	8	High	Madhanraj R K & Kishore samuvel S	Week 1
Sprint 2	Auth & Security	User schema, bcrypt hashing (10 salts), JWT signing, Bearer auth.js guard.	13	High	Devanand R	Week 2
Sprint 3	Workout CRUD Engine	Workout model, Add/View/Update/Delete workouts, date sorting, user scoping.	15	High	Abishek M	Week 3
Sprint 4	Search Engine	Multi-criteria regex search (?q=) across workoutName, category, and date.	10	High	Abishek M & Kishore samuvel S	Week 3-4
Sprint 5	Gemini AI Engine	Google Gemini SDK wrapper, prompt templates for workout plans & insights.	18	High	Madhanraj R K	Week 4-5
Sprint 6	QA & Middleware	Centralized errorHandler, Morgan logger, Postman test collection, latency testing.	12	High	Sunil kumar M	Week 5-6
Sprint 7	Demo & Deliverables	Video walkthrough, technical documentation, Naan Mudhalvan submission pack.	8	High	Full Team	Week 6

Sprint Governance & Risk Management:

- **Agile Velocity:** 84 total story points completed across 7 sprints with zero sprint rollover.
- **Quality Gate:** Every user story required passing Postman tests and code review before branch merging.
- **Dependency Strategy:** External Gemini API integrated via decoupled geminiService.js to isolate external API failures.